

(a) Structures, processes and compounds involved in photosynthesis in C4 plants include:

A – carbon dioxide

H – mesophyll cells

B – phosphorylation

I – guard cells

C – palisade cells

J – photorespiration

D – PEP carboxylase

K – photolysis

E – rubisco

L – oxaloacetate

F – malate

M – ATP synthase

G – RuBP

N – chlorophyll

Complete Table 7.1 by matching each description with **one** letter chosen from **A** to **N** to show the correct structure, process or compound.

You may use each number once, more than once or not at all.

Table 7.1

description	letter
enzyme with a high optimum temperature	
enzyme in bundle sheath cells	
process that slows down the rate of photosynthesis	
compound that releases carbon dioxide into bundle sheath cells	
cells that stop oxygen reaching bundle sheath cells	

[5]

(b) Some plants, such as rice, are adapted to grow when partially submerged in water.

Describe **and** explain the adaptations that allow rice plants to survive in waterlogged fields.

..... [3]

[Total: 8]